



Week 02 Task-06

Vulnerable Lab Setup Using Metasploitable 2/DVWA

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1) Executive Summary

This task is based on creating a secure isolated virtual environment for Penetration Testing. The goal of this task is to setup a vulnerable machine named Metasploitable2 was installed in oracle virtual box. This isolated environment serves as the best environment for all security assessment tasks like network scanning, vulnerability scanning, exploiting the vulnerabilities reported in machine.

2) Scope

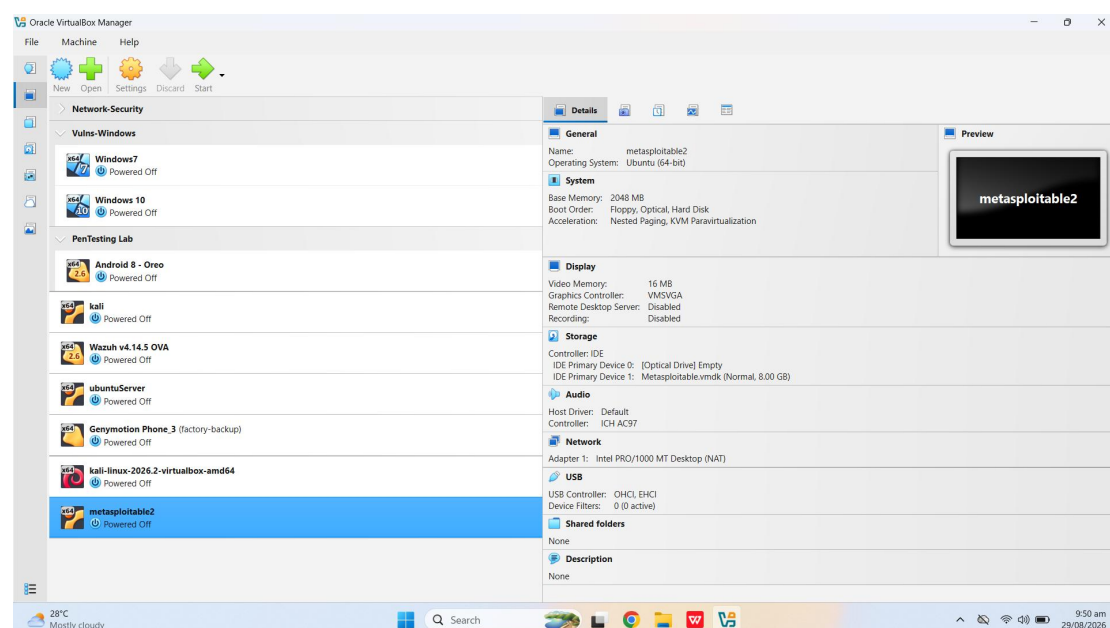
The scope of this task is to setup an environment using metasploitable2 in a virtual box, configure the target and attacker machine on the same network and verify the connectivity between attacker machine (**Kali Linux**) and target machine (**Metasploitable2**) for further penteting tasks.

3) Tools Used

1. Kali Linux
2. Oracle Virtual box
3. Metasploitable2
4. Windows 11 Host operating System

4) Metasploitable2 Setup

Metasploitable2 machine is downloaded from the Internet the website link is <https://www.rapid7.com/products/metasploit/metasploitable/> and it is imported in virtual box using the downloaded vmdk file. The machine was configured with appropriate hardware resources for security assessment.



```
Warning: Never expose this VM to an untrusted network!
Contact: msfdev[at]metasploit.com
Login with msfadmin/msfadmin to get started

metasploitable login: msfadmin
Password:
Last login: Sun May 20 15:50:42 EDT 2012 from 172.16.123.1 on pts/1
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ _
```

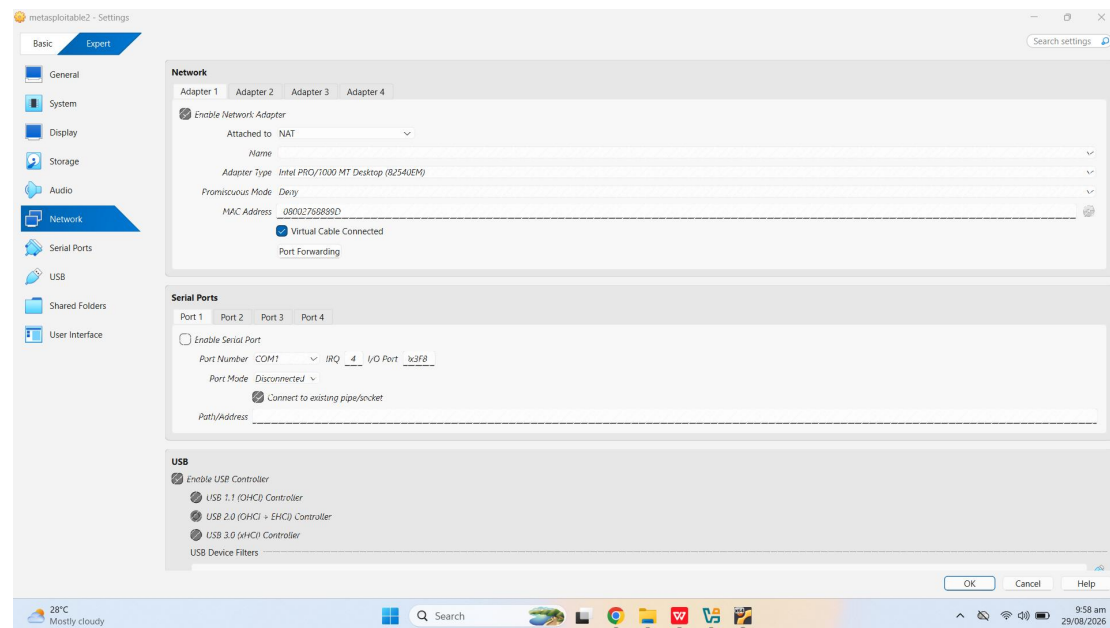
As in the above picture it is clearly shown that metasploitable 2 is successfully initiated.

Identification of target IP address

```
msfadmin@metasploitable:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 16436 qdisc noqueue
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 08:00:27:68:88:9d brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global eth0
    inet6 fd17:625c:f037:2:a00:27ff:fe68:889d/64 scope global dynamic
        valid_lft 86186sec preferred_lft 14186sec
    inet6 fe80::a00:27ff:fe68:889d/64 scope link
        valid_lft forever preferred_lft forever
msfadmin@metasploitable:~$ _
```

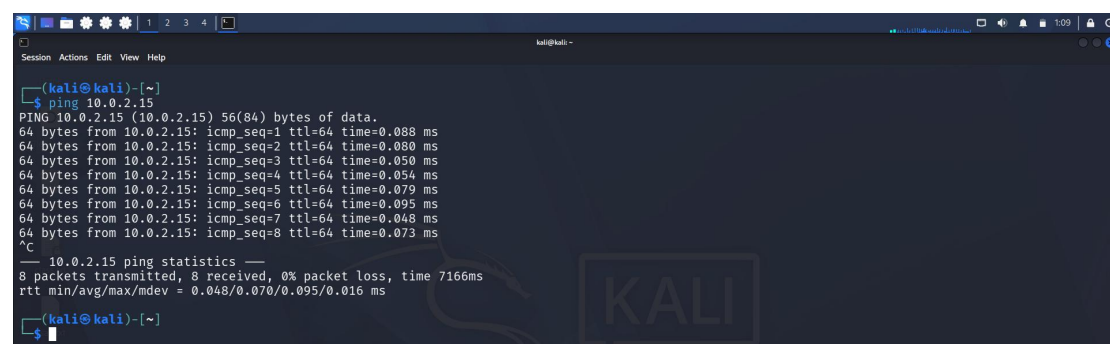
To identify the target IP address type command **ip addr** and the ip retrieved is 10.0.2.15/24 This is the IP address will be used for further penetration testing.

5) Network Configuration



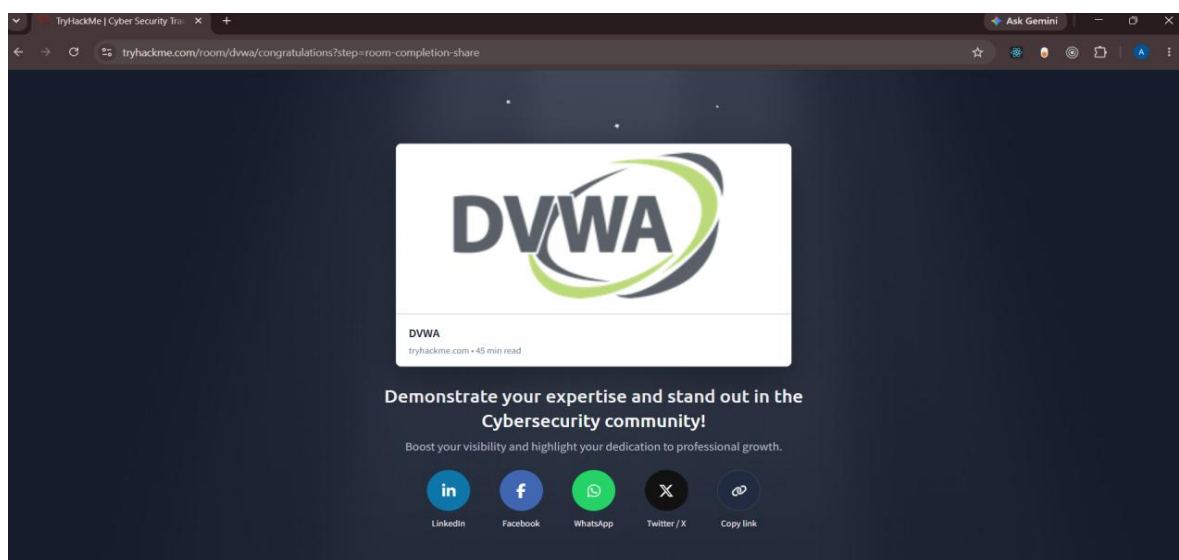
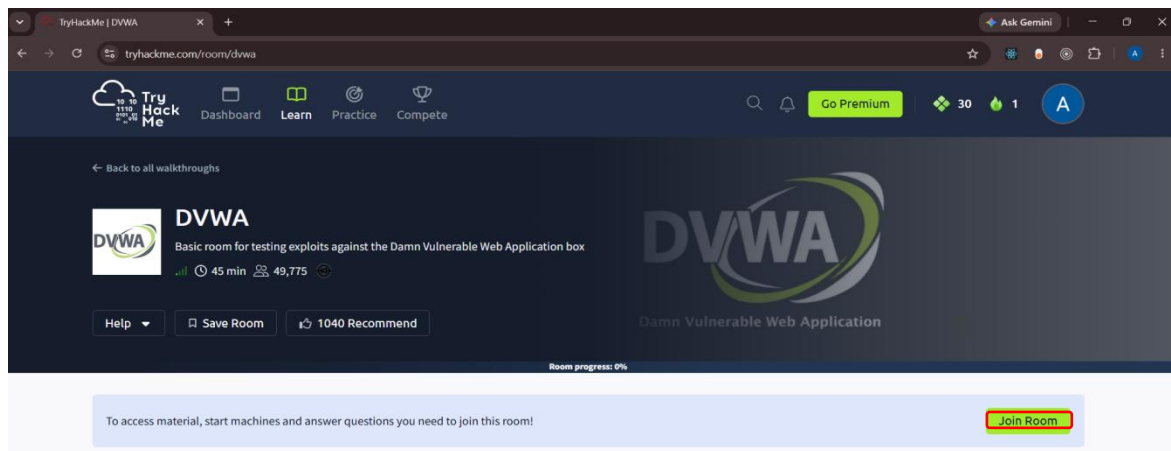
The attacker machine and the target machine both are configured on the same network adapter (NAT) in virtual box. This configuration allows both machines to communicate seamlessly as they are on the same network while maintaining the internet connection for downloading and updating tools.

6) Connectivity Testing



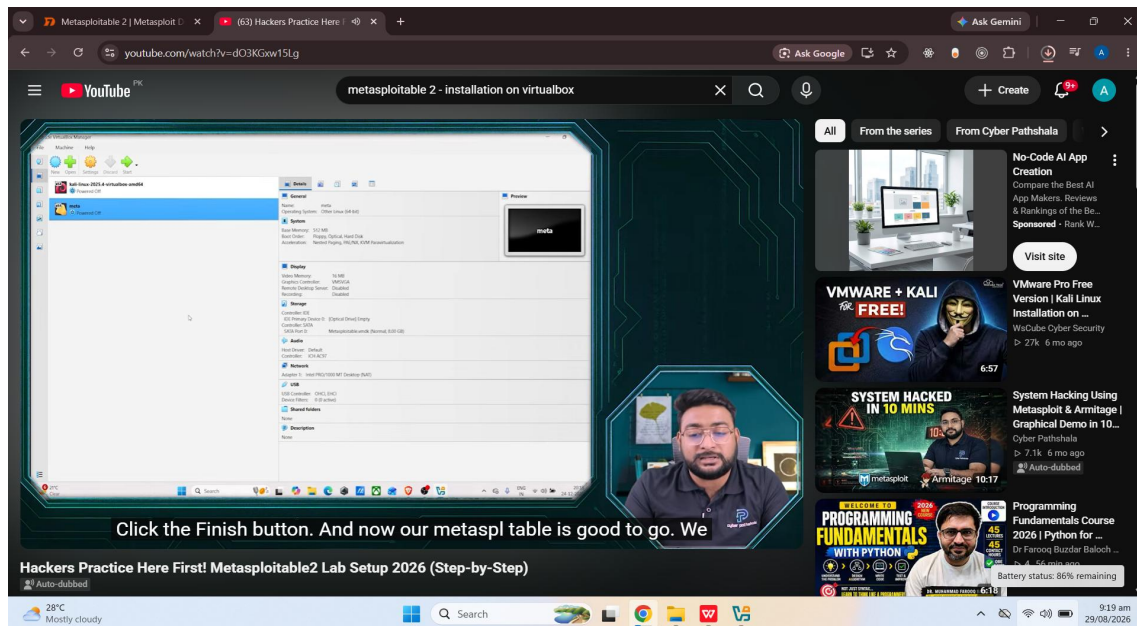
In the above figure it is clearly shown that all packets are transmitted and received 100% there is no packet loss this indicates that both attacker and target machine are connected means reachable.

7) TryHackMe DVWA Practical



The DVWA Try hack me room help me to understand setup and accessibility of **Damn Vulnerable Web Application (DVWA)**, This practical task provided hands-on experience of connecting with a vulnerable application.

8) Youtube video to download Metasploitable2



9) Conclusion

This task serves as the building ethical hacking lab for security assessment. In this task I created a secure environment using virtual box for pentesting tasks by deploying metasploitable 2 vulnerable machine.